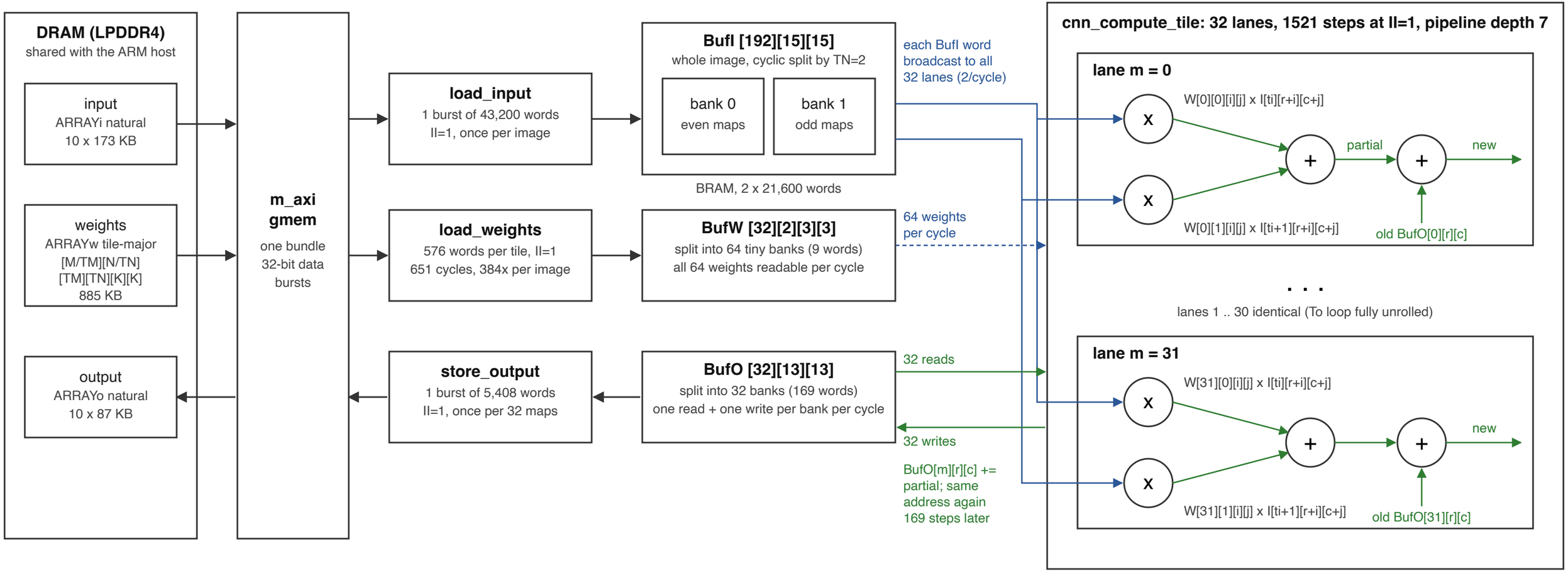
DRAM (LPDDR4)
shared with the ARM host
input
ARRAYi natural
10 x 173 KB
weights
ARRAYw tile-major
[M/TM][N/TN]
[TM][TN][K][K]
885 KB
output
ARRAYo natural
10 x 87 KB
m_axi
gmem
one bundle
32-bit data
bursts
load_input
1 burst of 43,200 words
II=1, once per image
load_weights
576 words per tile, II=1
651 cycles, 384x per image
store_output
1 burst of 5,408 words
II=1, once per 32 maps
BufI [192][15][15]
whole image, cyclic split by TN=2
bank 0
even maps
bank 1
odd maps
BRAM, 2 x 21,600 words
BufW [32][2][3][3]
split into 64 tiny banks (9 words)
all 64 weights readable per cycle
BufO [32][13][13]
split into 32 banks (169 words)
one read + one write per bank per cycle
each BufI word
broadcast to all
32 lanes (2/cycle)
64 weights
per cycle
32 reads
32 writes
BufO[m][r][c] +=
partial; same
address again
169 steps later
cnn_compute_tile: 32 lanes, 1521 steps at II=1, pipeline depth 7
lane m = 0
W[0][0][i][j] x I[ti][r+i][c+j]
W[0][1][i][j] x I[ti+1][r+i][c+j]
partial
new
old BufO[0][r][c]
. . .
lanes 1 .. 30 identical (To loop fully unrolled)
lane m = 31
W[31][0][i][j] x I[ti][r+i][c+j]
W[31][1][i][j] x I[ti+1][r+i][c+j]
new
old BufO[31][r][c]